

Escape Auction

Concept Document

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Pitch Line

A frantic race for freedom where only those who master the auctions will manage to accumulate enough energy to be the first to escape the prison.

Introduction

Escape Auction is a competitive platformer game for 2 to 4 players, centered on a frantic race to escape a high-security prison. Each player takes on the role of an inmate determined to be the first to reach the exit gate. However, escape is not just about speed and agility: it is crucial to gather enough energy to clear the path to freedom.

Each player's energy is represented by a bar that must be charged to enable escape. The main challenge lies in the lightning-fast auctions that occur every 3 seconds, where players compete by bidding coins to acquire energy. Everyone starts with 10 coins and can earn more each round based on dice roll results—an element that adds unpredictability and strategic depth.

By combining platforming action with real-time auction mechanics, Escape Auction delivers an intense and dynamic experience. Resource management, the ability to read opponents, and quick decision-making are as vital as movement skills. The first player to reach the prison gate with maximum energy wins, making every match a thrilling, fun, and highly competitive race.

Demographic Breakdown

Despite its seemingly simple controls, Escape Auction requires some strategic thinking and careful resource management. For these reasons, the game is recommended for ages 10 and up and is only available on PC.

Feature List

- Competitive matches for 2 to 4 players;
- Prison-themed setting;
- Energy bar that must be filled to escape;

- Dynamic real-time auctions every 3 seconds;
- Resource management system with a dedicated currency;
- Luck element through dice rolls to earn extra coins;
- Strategic decision-making and reading opponent behavior;
- Victory is secured by being the first to reach the gate with a fully charged energy bar;
- Intense, fun, and highly competitive gameplay.

Feature List Breakdown

- **Competitive 2-4 Player Matches:** Players race against each other to escape first, balancing strategy and luck.
- **Prison-Themed Setting:** Navigate a visually immersive prison.
- **Energy Bar Progression:** Fill your energy bar to escape; managing it is key to victory.
- **Real-Time Auctions:** Dynamic auctions occur every 3 seconds for resources and advantages.
- **Resource Management:** Use a dedicated currency to buy items, actions, or upgrades strategically.
- **Dice-Based Luck:** Roll dice to earn extra coins, introducing unpredictability and comeback potential.
- **Strategic Decision-Making:** Observe opponents and plan actions carefully to outmaneuver them.
- **Victory Condition:** First player to reach the gate wins.
- **Intense Competitive Gameplay:** Fast-paced, engaging, and designed for high replayability.

Formal Description

Game State:

The game state contains all relevant information at a given moment of the match:

- `characters[i].position`: position (x, y) of player i's character (indices 1–4).
- `gate[i].position.x`: horizontal position of the finish line for player i.
- `energies[i]`: current energy of player i (0–100).

- `money[i]`: current money of player `i` (integer ≥ 0).
- `dice[i]`: value of the last dice roll of player `i` (1–6).
- `ff[i]`: indicates whether player `i` has forfeited the auction (true/false).
- `auction_active`: indicates whether the auction is active (true/false).
- `auction_players`: list of players currently participating in the auction.
- `bids[i]`: value of the current bid of player `i`.
- `highest_bid`: value of the current highest bid.
- `current_player_index`: index of the current player in the `auction_players` vector.
- `auction_winner`: winner of the auction (if any).
- `time_left`: remaining time in the auction timer.
- `is_finished`: indicates whether the race has ended (true/false).
- `num_players`: number of active players (1–4).
- `race_winner_id`: ID of the player who won the race.

`S`={ positions, energies, money, dice, ff, auction_active, auction_players, bids, highest_bid, current_player_index, time_left, is_finished, num_players, race_winner_id}

Game View:

Elements visible in the game interface:

During gameplay:

- Real-time character positions (with animated sprites)
- Each player's energy bar
- Each player's money balance
- Dice roll results
- Finish gates
- Dynamic camera, following all players

During auctions:

- Auction panel
- Current player
- Highest current bid

- Available funds
- Buttons ('Next Player' and 'FF' – Fast Forward)
- Value selected in the SpinBox
- Timer (mm:ss format)

Game Setup:

At the start of the game:

- The visibility of all players is determined by the total number of participants (num_players).
- **Initial Energy:**
energies[1..4] = 100 (only for active players).
- **Initial Money:**
money[1..4] = 10 (only for active players).
- **Starting Position:** Defined in the scene layout.
- **Auction Settings:**
 auction_active = false
 auction_players = []
 bids = {1:0, 2:0, 3:0, 4:0}
 highest_bid = 0
- **Auction Timer:**
 auction_timer = 3 seconds
- **Race Status:**
 is_finished = false
 race_winner_id = null

S0={energies=100,money=10,auction_active=false,bids=0,highest_bid=0,is_finished=false}

Player Actions:

During the race:

- move_forward(id) — Executes automatically while energy remains.
- stop() — Triggered automatically when energy reaches 0.
- reach_goal(id) — Triggered upon reaching the finish line (gate).

During the auction:

- bid(value) — Places a bid higher than the current leading bid.

- `pass()` — The player forfeits their turn in the auction (FF).
- `next_player()` — Passes the turn to the next player.

Set of actions:

$A = \{\text{move}, \text{stop}, \text{bid}(x), \text{pass}, \text{next_player}\}$

Victory Conditions:

Victory:

A player wins when their character passes through the corresponding gate:

$\text{character}(i).x \Rightarrow \text{gate}(i).x \Rightarrow \text{winner} = i$

So:

- `is_finished = true`
- `GameState.race_winner_id = i`
- The scene changes to the results screen.

Winning the auction (partial benefit):

The player who makes the highest bid in the auction wins:

$\text{auction_winner} = \text{argmax}(\text{bids}[i])$

And receive:

- Energy restored: `energies[winner] = 100`
- Reduced money: `money[winner] = money[winner] - bid[winner]`

This gives a competitive advantage in the race, but it's not the ultimate victory in the game.

Progression of Play:

The gameplay loop unfolds as follows:

1. **Race Start:** All characters begin running automatically.
2. **Energy Drain:** Their energy continuously depletes while moving.
3. **Cyclic Event (Every 3 seconds):**
 - All players roll the die.
 - They earn currency based on the die's value.
 - An auction is triggered.
4. **Auction Phase:**

- Players place bids in turns.
- They may forfeit their turn.
- The highest bid wins and fully restores the winner's energy.

5. **Resume Race:** The race resumes immediately after the auction.

6. **Race End:** When a character reaches their designated gate:

- The game concludes.
- A winner is determined.
- The results screen is displayed.

Summary flow:

Roll Dice→Auction→Energy Boost→Running→Goal→End